

Aiming for the Goal: Contribution Dynamics of Crowdfunding[†]

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We study a dynamic contribution game where investors seek private benefits offered in exchange for contributions, and a single, publicly minded donor values project success. We show that donor contributions serve as costly signals that encourage socially productive contributions by investors who face a coordination problem. Investors and the donor prefer different equilibria, but all benefit in expectation from the donor's ability to dynamically signal his valuation. We explore various contexts in which our model can be applied and delve empirically into the case of Kickstarter. We calibrate our model and quantify the coordination benefits of dynamic signaling in counterfactuals. (JEL C73, D26, D82, G32, L26, M13)

Contribution games typically feature positive externalities that result in free riding and miscoordination. These two forces can lead to underprovision, which has been widely studied in the context of public goods. Although free riding may be alleviated if investors receive private benefits in exchange for their contributions, miscoordination can persist due to uncertainty about future contributions. In this paper, we show how the dynamic interaction between differently motivated participants—a publicly motivated contributor and privately motivated investors—can alleviate miscoordination and promote successful funding of a project.

We study a contribution game where randomly arriving investors decide whether to contribute to a project that succeeds only if enough contributions are made before a deadline. If the project succeeds, investors who pledged support receive a fixed, private benefit. A representative, long-lived donor values the public benefits, but his valuation of the project, akin to a budget, is his private information. The donor can

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contribute throughout the game, and his payoff is equal to his valuation less his total contribution. If the project fails, all contributions are returned, but investors bear an opportunity cost. Therefore, investors contribute only if they believe the project will succeed with a sufficiently high probability. The probability of success is shaped by investor beliefs about the donor's valuation and the commitments made by the other investors over time.

Such contribution games arise in many settings. Our leading application is reward-based crowdfunding, where entrepreneurs raise funds via a combination of donations and purchases. Another application is fixed-duration capital campaigns launched by nonprofits targeting large leadership donors and small privately motivated contributors. The forces of our model also manifest in settings in which a government strategically times its interventions (e.g., with subsidies) to encourage investment in a sunrise industry or a multinational firm needs to encourage suppliers to invest in nearby facilities.

Our analysis yields two key new insights. First, we show that the donor can alleviate the coordination risk faced by private investors by signaling his valuation of the project. We derive bounds on the effect of dynamic signaling on the probability of project success and establish that making the donor's valuation public would result in the lowest probability of success. Second, we show that there exists a trade-off between maximizing the donor's and investors' payoffs. The donor prefers the equilibrium that maximizes the probability of success. However, this equilibrium does not internalize investors' opportunity costs of forgoing the outside option, and, therefore, investors prefer equilibria with intermediate probabilities of success.

Formally, we first show that a perfect Bayesian equilibrium that maximizes the probability of success can be attained by a Markov equilibrium with an intuitive structure involving a dynamic, state-dependent donation threshold. Donating less than the threshold stops privately motivated investments irreversibly, causing the project to fail, while donating more squanders funds. Hence, all donors with a sufficiently high valuation donate just above the required threshold. For this reason, we call such equilibria pooling-threshold (PT) equilibria. The threshold decreases after a pledge but increases in the absence of one.

In the success-maximizing equilibrium, the threshold is such that the donor makes investors just indifferent between contributing to the campaign and selecting the outside option. We construct this equilibrium by induction on the number of additional investors necessary to achieve the goal if no more donations are made. To establish that this equilibrium maximizes the probability of success, we use an innovative proof approach that recasts the problem as a dynamic information design problem. We then show that the solution to this relaxed problem is attained by the constructed equilibrium.

We next characterize the unique outcome that minimizes the probability of success in any perfect Bayesian equilibrium. This outcome is also attained by a PT equilibrium, but the donation thresholds are so high that investors contribute even if they assume that no additional donations will be made. The success-minimizing equilibrium outcomes coincide with a game where the donor's valuation is public information (i.e., a game without signaling).

Campaign participants prefer different equilibria. The donor prefers the success-maximizing equilibrium. Investors want to reduce their coordination risk

associated with forgoing their outside option; therefore, investors prefer equilibria with probabilities of success strictly between the success-maximizing and success-minimizing equilibria.

Because our proof approach involves recasting the success-maximization problem as a general information design problem, we can also establish that no other mechanism involving donations, fixed investments, and observable contribution histories can achieve a higher probability of success. For example, the probability of success is lower if the donor is restricted to contribute only at the beginning or at the end of a campaign.

We consider several extensions that highlight the broader applicability of our findings. For example, we show that the success-maximizing equilibrium is still achievable in the presence of multiple donors that know each other's valuations of the project. We also consider an extension where the campaign is no longer all-or-nothing, the goalpost is uncertain, and investments have some scrap value. This extension can represent a government (donor) seeking to encourage investment by private firms in developing industries or a large, multinational firm establishing a new location, seeking potential suppliers to invest in nearby facilities. We highlight that the government (respectively, multinational firm) can best coordinate firms (respectively, suppliers) by withholding some resources initially. A single, initial large push to get investors to make commitments, or fully revealing the level of support of the government (respectively, multinational firm), does not maximize the probability of success.

Finally, we apply our framework to reward-based crowdfunding using data from the platform Kickstarter. On this platform, entrepreneurs initiate campaigns to raise funds to launch new products (called "rewards"). The platform allows both donations and investments (i.e., pledges to buy a product). These campaigns map well to our theoretical model. Our sample covers all campaigns launched between March 2017 and September 2018 on Kickstarter. A novel aspect of our dataset is that it distinguishes pledges for rewards from donations at a 12-hour frequency. We document that donations constitute 14 percent of all funds raised.

Our empirical analysis proceeds in two steps. First, we validate our modeling assumptions and provide descriptive evidence consistent with equilibrium predictions. For example, we show that pledges are consistent with the two distinct contribution incentives of our model. Whereas pledges occur throughout time, donations decline close to zero after a campaign succeeds, as our model predicts. Second, we calibrate our model to the data and conduct counterfactuals. Our results emphasize the crucial role dynamic signaling plays at facilitating coordination because there is a significant gap in the success rate between the success-maximizing (16 percent) and the success-minimizing equilibria (3 percent). We also simulate counterfactuals that restrict donations to occur at the start (or end) of the campaign. Although imposing this restriction tends to result in larger initial commitments by the donor, we find that the success rate is even lower than the one we estimate under the success-minimizing equilibrium with continuous donations.

Related Literature.—Our setting is related to public good contribution games (Bagnoli and Lipman 1989; Admati and Perry 1991; Fershtman and Nitzan 1991; Varian 1994), but participants receive private benefits, as in Alaei, Malekian, and

Mostagir (2016).¹ At the core of our model is a socially minded donor who can incentivize contributions, akin to the leadership giver in Andreoni (1998), who can help overcome free riding. Although our setting has no free riding, we show that signaling by the donor alleviates coordination risk.

Our results on success-maximizing mechanisms complement recent work examining optimal mechanism design in crowdfunding, including Strausz (2017), who finds the optimal mechanism in the presence of moral hazard of investors; Ellman and Hurkens (2019b), who find the optimal mechanism with price discrimination; and Chang (2020), who studies two funding mechanisms when the quality of the project is unknown.²

We also contribute to the literature on dynamic signaling. Unlike in classic dynamic signaling environments, including Noldeke and Van Damme (1990) and Swinkels (1999), in our setting investments are welfare enhancing and are not wasteful. Our construction of PT equilibria involves belief thresholds to guarantee participation, as is common in models of reputation, contests, and lemons markets (Bar-Isaac 2003; Daley and Green 2012; Gul and Pesendorfer 2012; Kolb 2019; Gryglewicz and Kolb 2022). Similar to Ely and Szydlowski (2020), we solve an information design problem of a principal with private information, but our setting entails multiple agents who face coordination problems themselves. As a result, our donation thresholds depend on a stochastically changing state, as in Gryglewicz and Kolb (2022). We adopt an approach of optimizing over beliefs rather than strategies, similar to that used in dynamic information design and mechanism design with limited commitment (e.g., Doval and Ely 2020; Doval and Skreta 2022).

Finally, we provide new empirical insights on reward-based crowdfunding. Our findings connect to the economics of innovation (Belleflamme, Lambert, and Schwenbacher 2014; Lee and Persson 2016; Sorenson et al. 2016; Grüner and Siemroth 2019) and complement experimental work on how contributions affect campaign success (Van de Rijt et al. 2014). We quantify the effects of prosocial incentives (Kuppuswamy and Bayus 2018; Dai and Zhang 2019), which have also been observed in equity crowdfunding (e.g., Agrawal, Catalini, and Goldfarb 2015) and capital campaigns (e.g., List and Lucking-Reiley 2002).

I. Model of Crowdfunding

In this section, we present our baseline model of a contribution game. The objective is to raise a goal amount $G > 0$ by a deadline $T > 0$. We say the project (also referred to as the “campaign”) succeeds if the goal amount is raised by the deadline. Time is divided into periods of length Δ . Let $\mathbb{T}^\Delta := \{\Delta, 2\Delta, \dots, T\}$ denote the set of periods. For any period $t \in \mathbb{T}^\Delta$, there is a corresponding time remaining $u := T - t \in \mathbb{U}^\Delta := \{T - \Delta, \dots, \Delta, 0\}$.

¹ See also Marx and Matthews (2000); Bonatti and Hörner (2011); Campbell, Ederer, and Spinnewijn (2014); Cvitanic and Georgiadis (2016); and Sahm (2020). Our paper is also related to work on crowdfunding, including Chakraborty and Swinney (2021); Chemla and Tinn (2020); Ellman and Fabi (2022); Chakraborty, Ma, and Swinney (2023).

² Ellman and Hurkens (2019a) relax the ex post individual rationality constraint. Belavina, Marinesi, and Tsoukalas (2020) further distinguish between funds misappropriation and performance opacity.

Players and Payoffs.—There are two types of contributors: randomly arriving investors (each referred to as “she”) and a single, representative long-lived donor (referred to as “he”). In each period, an investor arrives with probability $\Delta\lambda \in (0, 1)$. Upon arrival, she makes a one-shot decision to either pledge to pay p or choose an outside option of value $v_0 > 0$. If the project succeeds, all investors who pledged to pay p receive a reward of $v > 0$, resulting in a payoff of $v - p > v_0$. If the project fails, investors pay nothing and receive a payoff of zero. In the context of reward-based crowdfunding, v is an investor’s valuation of a product. The term v_0 can be interpreted as the value of a short-lived investment opportunity.³

The long-lived donor values a successful project at $w \geq 0$. If the project succeeds, the donor’s payoff is $w - D_T$, where D_T is the total contribution made by the donor. If the project fails, donations are returned, and the donor’s payoff is zero. Hence, the donor never donates more than w . Investors do not observe w . They only know that it is drawn from a distribution on $[0, \infty)$, with a continuously differentiable and strictly increasing cumulative distribution function (cdf) F_0 . We denote the random variable by W .

Timing, Histories, and Strategies.—Figure 1 illustrates the timing of the game. Within a period, if an investor arrives, she decides whether to pledge p or not. Then, the donor decides whether and how much to donate. The donor is also allowed to donate at the start of the game ($t = 0$). We denote cumulative pledges and cumulative donations up to and including period t by N_t and D_t , respectively. The final project revenue is equal to $R_T = D_T + N_T p$. Success requires $R_T \geq G$.

The history $h_t^{B,\Delta}$ of an investor who arrives in period t consists of all cumulative pledges and donations up to period $t - \Delta$. We denote the set of period- t investor histories by $\mathcal{H}_t^{B,\Delta}$.⁴ A period- t donor history $h_t^{D,\Delta}$ additionally includes the period- t pledge, if any. We denote the set of period- t donor histories by $\mathcal{H}_t^{D,\Delta}$. A strategy of a period- t investor is a mapping $\tilde{b}_t^\Delta: \mathcal{H}_t^{B,\Delta} \rightarrow [0, 1]$, where $\tilde{b}_t^\Delta(h_t^{B,\Delta})$ is the probability of an investor purchasing at history $h_t^{B,\Delta}$. We denote the collection of all investor strategies by $\tilde{b}^\Delta := (\tilde{b}_t^\Delta)_t$. The donor strategy is a mapping $\tilde{D}_+^\Delta: \bigcup_{t \in T^\Delta} \mathcal{H}_t^{D,\Delta} \times [0, \infty) \rightarrow \mathbb{R}$, where $\tilde{D}_+^\Delta(h_t^{D,\Delta}; w) = D_t \geq D_{t-\Delta}$ represents cumulative donations after the donation decision at history $h_t^{D,\Delta}$.⁵ Investor beliefs, $\tilde{F}^\Delta: \bigcup_{t \in T^\Delta} \mathcal{H}_t^{B,\Delta} \rightarrow [0, 1]^\mathbb{R}$, map each investor history $h_t^{B,\Delta}$ to a cdf $\tilde{F}^\Delta(\cdot; h_t^{B,\Delta})$.

Solution Concept.—A perfect Bayesian equilibrium (PBE) is given by an assessment $(\tilde{b}^\Delta, \tilde{D}_+^\Delta, \tilde{F}^\Delta)$, or a tuple of strategies and beliefs, such that

- the donor strategy $\tilde{D}_+^\Delta$ maximizes the donor’s expected payoff at any donor history $h_t^{D,\Delta}$ given investor strategies and beliefs;

³ It can also be interpreted as an inspection cost, a transaction cost of pledging, or a disappointment cost of not receiving a product if participants are loss averse.

⁴ We assume that investors do not observe arrivals but only pledges. Arrivals are not payoff relevant, so the Markov equilibria we characterize later are also equilibria in a game where arrivals are observed.

⁵ Formally, we allow for mixed strategies. In that case, $\tilde{D}_+^\Delta(h_t^{D,\Delta}; w)$ denotes the random variable that describes the mixed strategy at the corresponding history.

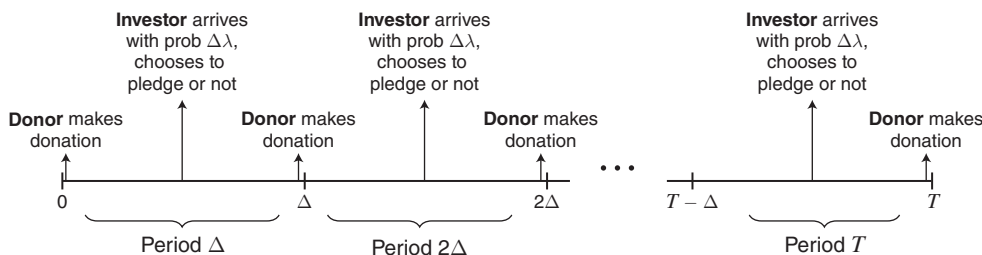


FIGURE 1. TIMING OF THE GAME

Notes: The donor can contribute at the beginning of time ($t = 0$). Then, within each period, an investor either arrives and then decides whether to pledge or does not arrive. The donor then decides if and how much to donate. The game ends at $t = T$.

- each period- t , investor strategy $\tilde{b}_t^\Delta$ maximizes the expected payoff of the investor at any history $h_t^{B,\Delta}$, given investor beliefs, and the donor and other investor strategies;
- investor beliefs about W , $\tilde{F}^\Delta(\cdot; h_t^{B,\Delta})$, are derived from all strategies according to Bayes' rule whenever possible.

An equilibrium outcome is given by a sequence of cumulative pledges and donations. The distribution of investor arrivals, the donor's valuation distribution, and an assessment $(\tilde{b}^\Delta, \tilde{D}_+^\Delta, \tilde{F}^\Delta)$ induce a probability measure $\mathbb{P}$ that governs equilibrium outcomes.

Payoff-Relevant State and Markov Equilibria.—All players' payoffs depend only on the cumulative number of pledges and the cumulative donation amount. Therefore, we define the payoff-relevant state in period t to be

$$\mathbf{x} := (N, D, u) \in \mathbb{X}^\Delta := \mathbb{N} \times [0, \infty) \times \mathbb{U}^\Delta,$$

or equal to $(N_{t-\Delta}, D_{t-\Delta}, T - t)$ for an investor, and $(N_t, D_{t-\Delta}, T - t)$ for a donor.

Donor strategies, investor strategies, and investor beliefs are said to be *Markovian* if they depend only on the state, both on and off the equilibrium path. These objects are represented by $D_+^\Delta : \mathbb{X}^\Delta \times [0, \infty) \rightarrow \mathbb{R}$, $b^\Delta : \mathbb{X}^\Delta \rightarrow [0, 1]$, and $F^\Delta : \mathbb{X}^\Delta \rightarrow [0, 1]^\mathbb{R}$, respectively. We call PBEs in Markovian strategies and beliefs *Markov equilibria*, described by a Markovian assessment $(b^\Delta, D_+^\Delta, F^\Delta)$.

II. Characterization of Equilibrium Outcomes

In this section, we characterize the game's PBE and bound the effects of dynamic signaling on the probability of success. We identify PBEs preferred by the donor and investors, respectively. Finally, we demonstrate that alternate campaign designs with varied donation timings or information about donor valuation cannot increase the probability of success. All proofs are in online Appendix B.

A. Preliminaries

Given a Markovian assessment $(b^\Delta, D_+^\Delta, F^\Delta)$, let $\pi^\Delta(\mathbf{x})$ denote the induced probability of success from the perspective of the $(N + 1)$ st investor if she pledges in state $\mathbf{x} = (N, D, u)$. Upon arrival, an investor in state $\mathbf{x}$ is willing to pledge if and only if the expected utility of pledging is greater than the utility of the outside option. That is,

$$(\text{Investor-PC}) \quad \pi^\Delta(\mathbf{x}) \cdot (v - p) \geq v_0.$$

The probability of success $\pi^\Delta(\cdot)$ is determined by the assessment $(b^\Delta, D_+^\Delta, F^\Delta)$ and two sources of uncertainty. First, investors are uncertain about the donor's valuation w and update their beliefs based on the observed state $\mathbf{x}$. Second, there is uncertainty about the number of future investor arrivals. We say a project is active in state $\mathbf{x}$ if beliefs are sufficiently high to incentivize pledging, that is, $\pi^\Delta(\mathbf{x}) \geq v_0/(v - p)$.

B. Pooling-Threshold Equilibrium Structure

Next, we describe Markov equilibria in which the donor strategies have an intuitive structure. We later establish that equilibrium outcomes that maximize or minimize the probability of success can be attained by such equilibria. In these equilibria, all donors with a sufficiently high valuation donate just above a state-dependent threshold (see Definition 1).

DEFINITION 1: We call a Markovian donor strategy D_+^Δ a pooling-threshold strategy if for any N and u , there is a donation threshold $D_*^\Delta(N, u) \geq 0$, with $D_*^\Delta(N, 0) = G - Np$, such that

$$D_+^\Delta(\mathbf{x}; w) = \max\{D, D_*^\Delta(N, u)\}, \quad \forall w \geq D_*^\Delta(N, u), \text{ and}$$

$$D_+^\Delta(\mathbf{x}; w) = \max\{w, D\}, \text{ otherwise.}$$

Given a PT donor strategy, donations serve to signal the donor's valuation. If donations are sufficiently high, investors are optimistic that the project will ultimately succeed. The donation threshold ensures that the next arriving investor will believe that the campaign succeeds with a probability higher than $v_0/(v - p)$. In equilibrium, investors pledge if and only if cumulative donations exceed the donation threshold of the donor in the preceding period,

$$(\text{PT-investor}) \quad b^\Delta(N, D, u) = 1 \Leftrightarrow D \geq D_*^\Delta(N, u + \Delta).$$

Additionally, for the Markov equilibria that we construct, the donation threshold $D_*^\Delta(N, u)$ is decreasing in both N and u . Thus, fewer donations are required to keep

the project active when more investors have pledged and more time remains until the deadline. As a result, these equilibria can be supported by investor beliefs that are truncations of the prior distribution F_0 when $D \geq D_*^\Delta(N, u + \Delta)$. PT beliefs are equal to

$$(\text{PT-belief}) \quad F^\Delta(w; \mathbf{x}) = \begin{cases} \frac{F_0(w) - F_0(D)}{1 - F_0(D)} \cdot \mathbf{1}\{w \geq D\}, & \text{if } D \geq D_*^\Delta(N, u + \Delta) \\ \mathbf{1}\{w \geq D\}, & \text{otherwise} \end{cases}.$$

As soon as cumulative donations fall below $D_*^\Delta(N, u + \Delta)$, investors believe that the donor has exhausted his valuation, and $w = D$.⁶ We define a PT assessment in Definition 2.

DEFINITION 2: An assessment $(b^\Delta, D_+^\Delta, F^\Delta)$ is a pooling-threshold assessment if D_+^Δ is a PT strategy with a donation threshold $D_*^\Delta(N, u) \in [0, G - (N + 1)p]$ for $u > 0$, if equations (PT-investor) and (PT-belief) are satisfied, and if the following conditions hold for $u > 0$:

(i) Weak monotonicity in u and strong monotonicity in N , that is,

$$D_*^\Delta(N, u) \geq D_*^\Delta(N + 1, u - \Delta) \geq D_*^\Delta(N + 1, u);$$

(ii) Strict monotonicity in N , that is, $D_*^\Delta(N, u) > D_*^\Delta(N + 1, u)$ if $D_*^\Delta(N, u) > 0$;

(iii) No donation threshold after success, that is, $D_*^\Delta(N, u) = 0$ for $(N + 1)p \geq G$.

Condition (i) in Definition 2 imposes that the donation threshold is weakly decreasing in u but also that if there is a pledge in period $u - \Delta$, then the donation threshold $D_*^\Delta(N + 1, u - \Delta)$ does not increase. The monotonicity in N is stronger than the monotonicity in u . Condition (ii) simply imposes strict monotonicity of D_*^Δ in N , and Condition (iii) requires that D_*^Δ drops to zero if investors alone raise the goal amount.

Because the donation threshold is decreasing in N and u , once a project becomes inactive, it can never be active again. This condition allows us to define cutoff times (CT) such that given realized donor valuation w and the number of additional investors j needed for success, the project will fail unless the next investor arrives before the cutoff time $u = \xi_j^\Delta(w)$. Let $\underline{M}(D) = \lceil \frac{G-D}{p} \rceil$ denote the total number

⁶Many other investor beliefs can sustain a PBE in which the donor plays a PT strategy. Technically, the beliefs chosen here violate the “cannot signal what you do not know” condition off the equilibrium path, as introduced in Fudenberg and Tirole (1991), in the sense that early investor pledges can affect the beliefs of later investors independent of the donor’s actions. We could recover the aforementioned condition without altering anything qualitatively by imposing that for any off-path history $h_t^{B,\Delta}$ such that there exists a $s \leq t$ with $D_s < D_*^\Delta(N, T - s)$, we have $F(w; h_t^{B,\Delta}) = \mathbf{1}\{w \geq \min\{D_s : D_s < D_*^\Delta(N, T - s), s \leq t\}\}$. Instead of allowing such non-Markovian, off-path beliefs, we choose the Markovian on- and off-path beliefs given in equation (PT-belief) for their clean structure.

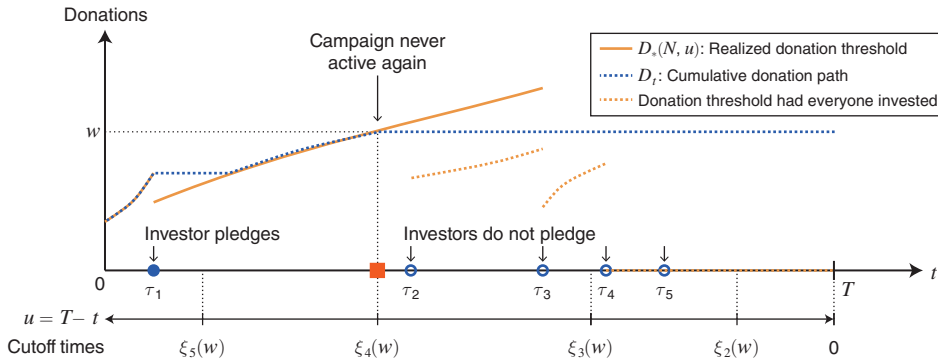


FIGURE 2. SAMPLE EQUILIBRIUM PATH GIVEN DONOR VALUATION w SUCH THAT $\underline{M}(w) = 5$

Notes: The bottom line shows the cutoff times $\xi_j(w)$ by which the $(\bar{M}(w) - j + 1)$ -th investor must arrive in order for the project to stay alive. The x-axis marks time and depicts realizations of investor arrivals (by blue dots). Also represented are the donation thresholds and cumulative donations (up to w). The second investor does not arrive in time; that is, $\tau_2 > T - \xi_4(w)$. The donor runs out of funds, and the project fails at $T - \xi_4(w)$. We also include a counterfactual donation threshold had everyone invested.

of investors needed for success given total donations D , if no further donations are made. Then, for $j \leq \underline{M}(w)$,

$$(CT) \quad \xi_j^\Delta(w) := \min \left\{ u \in \mathbb{U}^\Delta : \pi^\Delta(\underline{M}(w) - j, w, u) \geq \frac{v_0}{v - p} \right\},$$

where we set $\xi_j^\Delta(w) = T$ if $\pi^\Delta(\underline{M}(w) - j, w, u) < v_0/(v - p)$ for all $u \in \mathbb{U}^\Delta$. The monotonicity of D_*^Δ guarantees that $\pi^\Delta(\underline{M}(w) - j, w, u) \geq v_0/(v - p)$ for $u \geq \xi_j^\Delta(w)$. Therefore, given state $\mathbf{x}$ with $N = \underline{M}(w) - j$, a donor with valuation w is not able to satisfy equation (Investor-PC) if and only if $u < \xi_j^\Delta(w)$. Investors will pledge in equilibrium if and only if they arrive before the specified cutoff times. If they arrive too late, the donor will “run out of funds” because he would need to donate more than w to keep the project active. Ex ante, there is an information asymmetry between donor and investors about $\xi_j^\Delta(w)$, but once period $\xi_j^\Delta(w)$ is reached, the asymmetry is resolved.

We construct Markov equilibria in which the donor plays a PT strategy, which we refer to as *pooling-threshold equilibria*. To construct PT equilibria, we define a donation threshold $D_*(N, u)$ and show that the equilibrium conditions are satisfied. We show that for any PT assessment, beliefs are consistent with Bayes’ rule and donors are best-responding to the investor strategy. Hence, for any specific PT equilibrium, it remains to show only that investors are best-responding as well.

To provide some intuition, in Figure 2 we illustrate a PT equilibrium and a sample equilibrium path as $\Delta \rightarrow 0$. We have chosen w such that $\underline{M}(w) = 5$ investors are needed for success. The horizontal axis marks time. The τ s mark investor arrival times, and $\xi_j(w)$ s denote calculated cutoff times. The N th investor must arrive when at least $\xi_{\underline{M}(w)-N+1}(w)$ time remains before the deadline in order to be willing to pledge. If the N th investor arrives after that instant, the expected utility from pledging for this investor (and subsequent investors) drops below v_0 , and the project fails.

In Figure 2, the donor contributes along the donation threshold (solid orange line) through the first investor arrival at τ_1 . The donor does so to ensure that the first

investor pledges. After this investor pledges, the donation threshold drops, and the next arriving investor's belief that the project will succeed increases. As a result, donations stop for a while and later resume along the updated donation threshold to keep the project active. In the figure, the second investor arrives after $\xi_4(w)$, and the donor runs out of funds; that is, $D_t = w$ (dashed blue line). This means that all investors arriving after τ_2 choose the outside option, and the project fails. The dashed orange lines after $\xi_4(w)$ mark counterfactual donation thresholds if investors had arrived before the relevant cutoff times.

C. Success-Maximizing Equilibrium

We first construct a PBE that maximizes the probability of success. Because the donor is willing to donate up to his valuation if necessary, maximizing the probability of success boils down to providing investors with incentives to pledge whenever possible. In a success-maximizing equilibrium, even relatively low cumulative donations can be a sufficiently strong signal to make investors optimistic enough to pledge. Such optimistic beliefs can be generated by a PT strategy with a minimal donation threshold that can support a PBE. We calculate a minimum donation threshold for any given history and show that at this threshold, investors are exactly indifferent between pledging and not pledging. The donor should never donate more than this threshold because the threshold will increase in the future if investors do not pledge. Therefore, it is prudent for the donor to hold back funds to potentially induce later investors to pledge. All donors with a valuation greater than the threshold pool and donate just enough to meet the threshold. The following proposition summarizes the key properties of a success-maximizing equilibrium.

PROPOSITION 1 (Success-Maximizing Equilibrium):

- (i) *Given any $\Delta > 0$, there exists a success-maximizing PBE that is a PT equilibrium;*
- (ii) *Any sequence of these PT equilibria $\{(b^\Delta, D_+^\Delta, F^\Delta)\}_\Delta$ converges to a unique limit (b, D_+, F) as $\Delta \rightarrow 0$. The donor's limiting PT strategy admits a donation threshold $D_*(N, u) = \underline{D}(N, u)$, where $\underline{D}(N, u) = 0$ if $\pi(N, 0, u) > v_0/(v - p)$, and, otherwise, the next investor is made indifferent such that*

$$\pi(N, \underline{D}(N, u), u) = \frac{v_0}{v - p},$$

where $\pi := \lim_{\Delta \rightarrow 0} \pi^\Delta$ (the limit being uniform in D). Given a realized valuation w and cumulative pledges N with $\underline{M}(w) - N = j > 1$, the project fails in the limit at $\xi_j(w) := \lim_{\Delta \rightarrow 0} \xi_j^\Delta(w)$, satisfying

$$\pi(\underline{M}(w) - j, w, \xi_j(w)) = \frac{v_0}{v - p}.$$

If $j \leq 1$, the project never fails.

In the proof, we first construct a PT equilibrium for fixed Δ that satisfies the analogous discrete-time properties described in Proposition 1. We then show that this PBE maximizes the probability of success and take the limit as $\Delta \rightarrow 0$.

The construction uses the insights provided in Section IIB. We cannot directly define a donation threshold for every state such that investors are indifferent between pledging and not pledging because such a condition is endogenous to all participants' strategies. Instead, we use an induction in the number of investors needed to reach the goal if no additional donations are made, $j = \underline{M}(D) - N$. Within each induction step in j , we construct equilibrium objects $D_*^\Delta(N, u)$, $\pi^\Delta(N, D, u)$, and $\xi_j^\Delta(w)$.

A success-maximizing PBE is simply one that maximizes investor pledges. The exact donation amounts do not matter, as long as they incentivize pledging. It suffices to consider "reduced histories" that ignore the precise donation amounts and keep track only of whether a donation keeps the project active. We recast the problem by directly choosing probabilities of reaching each reduced history subject to a martingale constraint and an investor participation constraint. We show that the PBE constructed in the first step achieves the value of this relaxed problem.

D. Success-Minimizing Equilibrium: No Signaling Benchmark

In the unique success-minimizing equilibrium, the ability of the donor to signal his valuation does not facilitate coordination between investors. The donor plays a PT strategy with a donation threshold that is so high that investors are willing to pledge even if the donor stopped contributing. To define this threshold, suppose that w is known to be zero, that is, $F_0(w) = \mathbf{1}\{w \geq 0\}$, and let $\bar{\xi}_j^\Delta := \xi_j^\Delta(0)$ be the cutoff times from Proposition 1. Furthermore, define $\bar{\xi}_j := \lim_{\Delta \rightarrow 0} \bar{\xi}_j^\Delta$, $\bar{D}^\Delta(N, u + \Delta) := \max\{G - (j - 1)p - Np, 0\}$ for $u \in (\bar{\xi}_{j-1}^\Delta, \bar{\xi}_j^\Delta]$, and

$$(\bar{D}) \quad \bar{D}(N, u) := \max\{G - (j - 1)p - Np, 0\} \text{ for } u \in (\bar{\xi}_{j-1}, \bar{\xi}_j] \text{ as } \Delta \rightarrow 0.$$

This is the maximum donation threshold that can arise in a PBE. Indeed, in any PBE, the donor would not contribute if total revenue already exceeded $G - (j - 1)p$ at $u \in (\bar{\xi}_{j-1}^\Delta, \bar{\xi}_j^\Delta]$ because all future investors would pledge anyway. As $\Delta \rightarrow 0$, donors always donate exactly p in order to "compensate" for the absence of an investor arrival.

We show that the PT equilibrium with donation threshold $\bar{D}$ minimizes the probability of success (see Proposition 2). However, now, unlike in the success-maximizing PT equilibrium, when an investor pledges, she has a strict incentive to do so.

PROPOSITION 2 (Success-Minimizing Equilibrium):

- (i) Given any $\Delta > 0$, there is a unique success-minimizing PBE that is a PT equilibrium.
- (ii) Any sequence of such PT equilibria $\{(b^\Delta, D_+^\Delta, F^\Delta)\}_\Delta$ converges to a unique limit (b, D_+, F) as $\Delta \rightarrow 0$. The donor's limiting PT strategy has a donation threshold $D_*(N, u) = \bar{D}(N, u)$ given by equation $(\bar{D})$.

Note that if the donation threshold is $\bar{D}$, then investors contribute as if the donor's valuation were equal to the current level of cumulative donations. As a result, in

this equilibrium, there is no signaling to aid coordination. The donor just serves to decrease the effective goal amount from G to $G - w$.

E. Donor- and Investor-Preferred Equilibria

Next, we investigate which equilibria are preferred by campaign participants. We show that there is a trade-off between maximizing donor and investor payoffs, but both parties benefit from the coordination arising from the donor signaling his valuation.

First, consider the donor-preferred equilibrium. If the donor could be refunded excess donations beyond the goal amount after the deadline (see Section III), then the donor would want to simply maximize the probability of success, and the success-maximizing equilibrium of Proposition 1 is also the donor's preferred equilibrium. When this is not possible, one might conjecture that the donor "overdonates" in the success-maximizing equilibrium and can benefit from reducing donations at the expense of a lower success probability. We show that this is not the case.

PROPOSITION 3 (Donor-Preferred Equilibrium): *The success-maximizing PT equilibrium outcome constructed in Proposition 1 is also optimal for the donor.*

The key step to proving Proposition 3 is to show that if there is a donor-preferred equilibrium, then there must be a donor-preferred equilibrium in which after any history, the donor donates either nothing or just enough to induce the next investor to pledge. The donor's problem then reduces to one in which he chooses probabilities over the same reduced histories (as in the Proof of Proposition 1) and minimal donation thresholds that satisfy a donor incentive compatibility constraint. Because early-arriving investors can always be induced to pledge with fewer cumulative donations than later-arriving investors, the donor-preferred equilibrium coincides with the success-maximizing equilibrium.

Next, we show that investors prefer equilibria with a probability of success strictly between the success-maximizing and success-minimizing equilibrium outcomes.

PROPOSITION 4 (Investor-Preferred Equilibrium): *For sufficiently small Δ , any PT equilibrium preferred by investors yields a probability of success strictly between the success-maximizing one in Proposition 1 and the success-minimizing one in Proposition 2.*

The success-maximizing equilibrium is not investor preferred because it does not internalize the opportunity cost of pledging. For example, suppose that the first investor arrives late in the game (with u time remaining) and observes total donation $\underline{D}^\Delta(0, u)$. This investor pledges in equilibrium, but if the donor type is such that he is nearly out of funds, then the probability of success is actually low, and pledging would be a mistake ex post. The investor would be better-off in a PT equilibrium with a slightly higher threshold where she could screen out such donor types who are nearly out of funds.

The success-minimizing equilibrium is also not investor preferred because investors benefit from some dynamic signaling. To understand why, suppose the realized

donor valuation is $w = G - p - \epsilon$ for small $\epsilon > 0$. Then, the project succeeds with two investors. The first investor contributes if $(v - p)[1 - (1 - \Delta\lambda)^{u/\Delta}] - v_0 \geq 0$, but total investor surplus is maximized if she instead considered

$$(v - p) \underbrace{\left[1 - (1 - \Delta\lambda)^{u/\Delta}\right]}_{\text{probability of at least one more arrival}} - v_0 + \underbrace{(v - p - v_0)\lambda u}_{\text{externality on future investors}} \geq 0.$$

Hence, incentivizing the first investor to contribute beyond $\bar{\xi}_2^\Delta(G - 2p)$ increases overall investor surplus. This can be achieved by lowering the donation threshold $\bar{D}^\Delta(0, u) = G - p$ on $u \in [\bar{u}, \bar{\xi}_2^\Delta(G - 2p))$ by ϵ , where $\bar{u}$ solves the above inequality with equality. We show formally that this lower donation threshold defines a PT equilibrium that investors prefer to the success-minimizing equilibrium.

F. Alternative Campaign Designs

We have shown that with incomplete information about the donor valuation, if the donor can dynamically signal his valuation over time, this can maximize the probability of success. A natural question to ask is whether there are campaign designs that outperform this one. The Proof of Proposition 1 offers some answers. Restricting the donor's ability to donate, or altering the information (revealing more or less at different times) about the donor's valuation, cannot increase the probability of success relative to the success-maximizing equilibrium outcome. This means that a campaign design in which donations are collected in a first stage and investors engage in a crowdfunding campaign in a second stage must result in a lower probability of success compared to the success-maximizing equilibrium outcome. The same is true when donations occur only at the deadline to "top up" the campaign. It is possible, however, that these alternative campaign designs result in a higher probability of success than that achieved under the success-minimizing equilibrium. We empirically investigate this possibility in Section IV.

III. Extensions

Next, we discuss some extensions that show that our central insights extend to a broad class of contribution games. Examples for this section can be found in the online Appendix.

A. Relaxing Modeling Assumptions

In our first set of extensions, we consider relaxing a single assumption at a time. We note any substantive changes to Propositions 1–4.

Multiple Donors: Although in our baseline model we consider a single donor, in order to avoid modeling the free rider problem, we argue that the success-maximizing equilibrium can still be supported when multiple donors observe each other's valuations.

To see why, assume that there are two (or any finite number of) long-lived donors with valuations w_1, w_2 such that $w := w_1 + w_2$ is distributed according to F_0 . Donors simultaneously decide contributions at the end of each period. The success-maximizing equilibrium can be supported by donor strategies in which the first donor contributes according to the PT strategy of Proposition 1 and the second donor does not contribute until the donation threshold reaches w_1 . Investor beliefs and strategies coincide with those of Proposition 1. However, we note that the presence of free riding can lower the probability of success in the success-minimizing equilibrium. For example, consider a two-period version of our model with $2p < w_1 < G < w_1 + w_2$. Both donors must contribute for the project to succeed, but it is also an equilibrium for both donors to not contribute.

Time-Varying Arrivals and Outside Options: For expositional reasons, in our baseline model we assume a constant arrival rate and outside option. Both assumptions can be relaxed. Our analysis remains unchanged with a time-varying arrival process as long as the arrival rate is smooth in t . A time-dependent outside option is also possible as long as it does not decrease too quickly. Otherwise, an inactive campaign can become active again at a later point in time, and the cutoffs $\xi_j^\Delta(w)$ may not exist. This would complicate the equilibrium analysis significantly.

B. Application: Crowdfunding

We capture additional features of crowdfunding with two natural extensions.

Project Creator as the Donor: Our exposition suggests that the campaign creator is distinct from the donor, but all results hold if the donor contributes to his own campaign and excess donations are returned at the deadline. This is because the probability of success and investor outcomes remain unchanged if donations are returned. Consequently, the bounds on the probability of success remain unchanged; however, additional equilibria may arise.

Social Learning about Product Quality: Our framework can accommodate environments in which investors learn about the quality of the product from other investor pledges. This scenario may be relevant when investment occurs prior to the production of a product.

To model social learning, we assume that investor valuations are vq , where v is deterministic and $q \in \{0, 1\}$ is the unknown quality of the project. Each investor receives an independent signal about quality $s \in \{0, 1\}$ that is equal to 1 with certainty if $q = 1$ and with a probability less than 1 if $q = 0$. Thus, investors who see $s = 0$ learn with certainty that $q = 0$ and do not pledge. The participation constraint of investors with a positive signal $s = 1$ is

$$\underbrace{\mathbb{P}(N_T p + D_T \geq G) | \mathbf{x}}_{\text{probability of success in state } \mathbf{x}} \left(\underbrace{\frac{E[(vq - p) \cdot \mathbf{1}\{N_T p + D_T \geq G\} | \mathbf{x}]}{\mathbb{P}(N_T p + D_T \geq G) | \mathbf{x}}}_{\text{expected value conditional on success}} - p \right) \geq v_0.$$

This expression shows that with social learning, donations can simultaneously increase the probability of success while lowering the expected benefits of contributing. This can make signaling equilibria possible but harder to sustain. We formalize this intuition in a two-period example in the online Appendix by showing that a PT equilibrium construction analogous to that in Proposition 1 is possible if there is not too much uncertainty about quality.

C. Application: Industrial Policy

Our framework can be applied to settings that are not all-or-nothing. Consider a government interacting with firms in a new sector, such as green energy. Typically, a government encourages investments in a sunrise industry by offering subsidies and other support. Firms receive private benefits for making an investment but benefit more if a critical mass of firms make complementary investments. We relax three assumptions and showcase how our framework can be used to study industrial policy in this context.

Alternative Investor Payoffs: To capture the notion of increased returns when contributions are complementary, we assume that if a firm makes an irreversible investment of p , this guarantees a return of at least v_L . If the sunrise industry takes off (i.e., the campaign is successful), investing firms get a higher payoff $v_H > v_L$. This results in an alternative investor participation constraint of the form

$$v_H \pi(N, D, u) + v_L (1 - \pi(N, D, u)) - p \geq v_0 \Leftrightarrow \pi(N, D, u) \geq \frac{v_0 - v_L + p}{v_H - v_L},$$

which is isomorphic to equation (Investor-PC) as long as $(v_0 - v_L + p)/(v_H - v_L) \in (0, 1)$. That is, with this assumption, our model immediately extends to campaigns that are not all-or-nothing.

Adding Uncertainty to the Goal: In new industries, there may be uncertainty ex ante about what defines success, and both the government and contributing firms may update their beliefs on the level of investment required for success. Formally, we assume that all participants face symmetric uncertainty about the goal, given by $\tilde{G} = G - X$, where $G, X \geq 0$ and $X \sim H$ on $[0, G]$; X is unknown to all players until contributions reach $\tilde{G}$. The government has the same payoff function as in our baseline model. Its valuation, or the fixed budget to spend, is $W \geq 0$, with $W \sim F_0$. The budget includes things that are not observed by firms (e.g., political capital).

This game is similar to our baseline model except that the campaign can succeed early if $D + Np + X \geq G$. We argue that the probability of success of a game where W is announced up front is lower than the probability of success of a game where W is the government's private information. That is, dynamic signaling is still valuable even if the goal is uncertain. The same argument holds if the government maximizes the probability of success subject to a private budget constraint, which captures the notion of earmarking.

Alternative Government Payoff: Investments made by the government may have some scrap value if the industry does not take off. Specifically, we consider the

government's payoff function to be $(W - \gamma D_T)\mathbf{1}\{R_T \geq G\} - (1 - \gamma)D_T$, where $\gamma \in [0, 1]$ is the scrap value of investment if the goal is not reached. Note that when $\gamma = 1$, the payoff function coincides with our baseline model.

The ability for government investments to be put to alternative uses if the campaign fails complicates the solution of the game because the probability of success affects the government's (donor's) decisions. The government may prefer to not contribute if success is unlikely because funds can be put to alternative uses. As a result, the donor-preferred equilibrium will no longer be success maximizing. Nonetheless, the qualitative forces remain: the ability for the government to dynamically signal its valuation can increase the probability of success. We illustrate this qualitative force using a two-period example in the online Appendix. The policy implication is that a government encouraging firms to make investments in a sunrise industry may be better-off by not offering an initial big push but rather providing incremental support as firms make commitments.

Alternative Interpretation: The game just described extends to other settings as well. The donor could represent a large multinational firm seeking to expand production in a new location. The firm needs to encourage a potential supplier network to provide components and services. Our model suggests that the multinational firm should dynamically respond to investments made by local complementary businesses.

IV. Empirical Application to Reward-Based Crowdfunding

We consider an empirical application to reward-based crowdfunding with two objectives. Our first objective is to document empirical patterns that are consistent with our model. Our second objective is to quantify the coordination benefits of dynamic signaling by calibrating our model and performing counterfactuals.

A. Reward-Based Crowdfunding Platforms

The core features of our model are similar to what is observed on reward-based crowdfunding platforms, including Kickstarter and Indiegogo. On these platforms, entrepreneurs raise funds to create new products. The entrepreneur specifies a funding goal (G), a funding deadline (T), and prices for rewards (p). Once a campaign goes live, these campaign features cannot be changed. Rewards are exclusive to the participants that pledge support to the campaign. Donating without receiving a reward is also possible.⁷ We refer to investors in this section as “buyers” and retain the term “donor” to refer to an individual who contributes without selecting a reward.

Most platforms use an all-or-nothing model so that transactions are realized if and only if the funding goal is reached by the deadline. Entrepreneurs may offer different reward versions; however, we abstract away from the use of price discrimination. After the campaign period, some rewards eventually become mass-market products,

⁷In principle, a participant can pledge for a reward and donate simultaneously by entering an amount greater than the reward price; however, we cannot observe this possibility without individual-level data.

TABLE 1—SUMMARY STATISTICS FOR THE DATA SAMPLE

Variable	Mean			Median	Fifth percentile	Ninety-fifth percentile
	(All)	(Uns.)	(Suc.)			
Campaign length	33.1	34.7	30.7	30.0	15.0	60.0
Goal (\$)	16,366.5	21,982.5	8,613.9	5,113.6	312.0	60,247.9
Number of rewards	7.2	6.0	9.3	6.0	1.0	17.0
Donor revenue (per 12 hours)	19.1	3.3	43.7	0.0	0.0	44.0
Buyer revenue (per 12 hours)	119.5	14.0	283.8	0.0	0.0	391.0
Total donations/total rev. (%)	29.2	33.1	24.6	15.8	0.0	100.0
Number of campaigns	42,462	24,624	17,838	—	—	—

Notes: Summary statistics for the 42,462 campaigns in the sample. Donor and investor contributions are the average over 12-hour periods. Means for all statistics are computed for all campaigns (All), unsuccessful campaigns (Uns.), and successful campaigns (Suc.). Also reported are the fiftieth, fifth, and ninety-fifth percentiles.

adding (social) value beyond the campaign. Examples of mass-market products that started as crowdfunding campaigns include the Peloton bike and Oculus VR headset (acquired by Meta). In line with our model, we do not examine postcampaign outcomes.

B. Data Sample and Summary Analysis

Our empirical analysis uses Kickstarter data for campaigns launched between March 2017 and September 2018. We create a panel by tracking every campaign, every 12 hours, from campaign launch through its deadline. Our data differentiate pledges for rewards from donations.⁸ This allows us to study campaign dynamics and interactions between participant types. In online Appendix D, we provide additional details on constructing our sample. We also perform 12 sets of robustness exercises, including treating some rewards (e.g., a thank-you card or meeting the entrepreneur) as a donation.

In Table 1 we provide summary statistics for the 42,462 campaigns included in our sample.⁹ Overall, 42 percent of campaigns succeed. Campaigns with larger goals tend to have longer deadlines (corr. coef. = 0.14). Unsuccessful campaigns tend to have a larger fraction of revenues coming from donations. Regardless of campaign outcome, donations are a key component of overall contributions, constituting 14 percent of total revenue.¹⁰

We plot additional campaign outcome summaries in Figure 3. In panel A, we show total revenue raised relative to the goal amount (R/G). There is considerable bunching at zero and one, meaning that most campaigns either receive little support or raise exactly the goal amount. In panel B, we plot the time period in which successful campaigns reach their goal. We refer to this time as the *success time*. Success

⁸Kuppuswamy and Bayus (2018) identify “family” contributions using last name matching based on self-selected usernames that were visible before 2012.

⁹Table D1 in online Appendix D contains summary statistics for the largest four categories.

¹⁰We maintain the assumption that rewards are exclusive but nonrivalrous because while campaign creators can assign capacity limits to rewards, these tend to be nonbinding. We find that 92 percent of rewards do not have a capacity limit, and only 21 percent of rewards with capacity limits ever sell out.

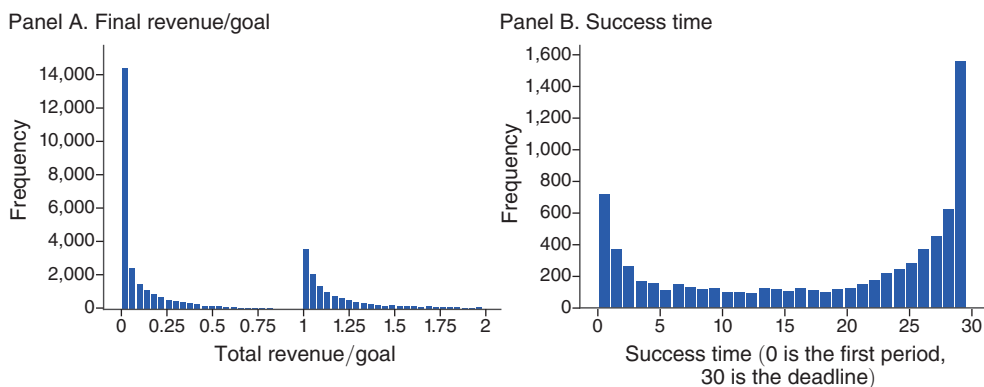


FIGURE 3. FREQUENCY HISTOGRAMS OF CAMPAIGN OUTCOMES

Notes: Panel A: Histogram of final campaign revenue (R) over the goal amount (G). The long, thin tail after $R/G = 2$ is not plotted. There is considerable bunching at 0 and 1. Panel B: Histogram of when 30-day campaigns succeed. Period $t = 0$ corresponds to the first day of the campaign, and $t = 30$ corresponds to the deadline. Success times are bimodal.

times are bimodal, with many campaigns succeeding either close to the start or close to the deadline.

C. Validating Model Assumptions and Theoretical Predictions

Before calibrating our model, we provide descriptive evidence on contributor incentives and contribution dynamics. Recall that in our theoretical model, the donor contributes solely to increase the probability of success, and buyers are motivated by obtaining private rewards. This suggests that donations should stop after a campaign reaches success, but buyer activity should continue. We estimate that donations drop by 72 percent within 3 days after the goal of a campaign is reached, whereas buyer contributions drop by only 37 percent.

Another salient feature of the data that suggests different contribution incentives is the strong relationship between the relative importance of donations versus purchases and success times. At the aggregate level, campaigns that succeed close to campaign launch should receive lower total donations than those that achieve their goals close to the deadline. To test this hypothesis, we partition the sample according to success times using three groupings: those that achieve the funding goal in the first 3 days (*early finishers*), those that achieve the goal in 3–27 days (*middle finishers*), and those that achieve the goal in the last 3 days before the deadline (*late finishers*). Table 2 provides summary statistics for these groupings that show stark differences between early- and late-finishing campaigns consistent with participants having different incentives. For example, the relative importance of donations increases in success time: 5 percent, 14 percent, and 34 percent for early, middle, and late finishers, respectively. In Figure 4, we plot contribution flows from buyers (panel A) and donors (panel B) for early-, middle- and late-finishing campaigns.¹¹

¹¹ We provide figures that measure the percentage of campaigns (also weighted by within-campaign revenues) and include unsuccessful campaigns in Figure D1 in online Appendix D.

TABLE 2—DESCRIPTIVE STATISTICS FOR EARLY-, MIDDLE-, AND LATE-FINISHING CAMPAIGNS

Variable	Early	Middle	Late
Goal (\$)	2,000.0	3,510.4	5,253.6
Number of rewards	8.0	8.0	8.0
Average price	90.6	118.7	171.5
R/G	3.1	1.2	1.0
$D/R(\%)$	4.8	14.7	34.3
$D/G(\%)$	17.0	20.9	36.7
Number of campaigns	1,437	3,470	2,616
Top categories	Theater, design, games	Theater, music, design	Theater, film and video, music

Notes: Summary statistics for successful campaigns partitioned by success time. Only 30-day campaigns are included. Early finishers are the campaigns that achieve the goal within three days. Late finishers are the campaigns that achieve the goal in the last three days. The middle category denotes campaigns that achieve the goal in 3 to 27 days. In this table, R denotes final revenue, G denotes the goal, and D denotes total donations. Medians reported. Donations become increasingly important for campaigns that finish late.

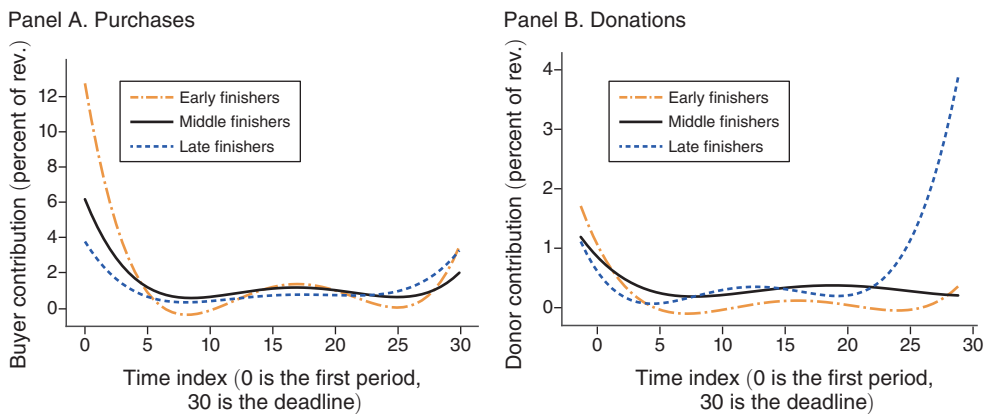


FIGURE 4. CONTRIBUTIONS OF BUYERS AND DONORS OVER TIME FOR SUCCESSFUL CAMPAIGNS

Notes: Panel A: Fitted values of a polynomial regression of degree 4 of time on percentage of total buyer revenues obtained within the period (B_t/R_T) . Panel B: Fitted values of a polynomial regression of degree 4 of time on percentage of total donor revenues obtained within the period (D_t/R_T) . For both panels, the three lines correspond to early finishers, middle finishers, and late finishers. Only 30-day campaigns are included. There is a spike in donations for late-finishing campaigns only.

While all campaigns have an initial spike in purchases at the beginning, the only spike for donations is for campaigns that succeed at the deadline.

Next, we establish properties on contribution dynamics of pooling-threshold equilibria and provide simple empirical evidence to validate these predictions in the data.

PROPOSITION 5 (Contribution Dynamics in Pooling-Threshold Equilibria): *All PT equilibria satisfy the following properties:*

- (i) *Campaigns that succeed at the deadline would fail without a donation and raise exactly the target amount with high probability. Formally, $\mathbb{P}(N_T p + D_{T-\Delta} < G | \mathcal{S}_T) > 1 - \Delta\lambda$ and $\mathbb{P}(D_T = G - N_T p | \mathcal{S}_T) \geq 1 - \Delta\lambda$, where $\mathcal{S}_t = \{N_{t-\Delta} p + D_{t-\Delta} < G \text{ and } N_t p + D_t \geq G\}$.*

- (ii) *Campaigns that succeed before the deadline succeed due to a buyer pledge. Formally, $\mathbb{P}(D_{\tau-\Delta} + N_{\tau}p \geq G) = 1$ if $\tau < T$.*
- (iii) *Donations drop to zero after a buyer pledges. Formally, $D_*^{\Delta}(N, u + \Delta) \geq D_*^{\Delta}(N + 1, u)$.*
- (iv) *Conditional on failing, campaigns with larger donor valuations fail later than campaigns with smaller donor valuations. Formally, given donor realizations $w > w'$, if a campaign is unsuccessful for both w and w' , and given the same buyer arrival realization, then the failure time $\iota := \inf\{t \geq 0 \mid \pi(N_t, D_t, T - t) < \frac{v_0}{v - p}\}$ is larger for w than for w' .*
- (v) *In success-minimizing PT equilibria, all donations are at least p .*

Empirical Evidence (Proposition 5)

- (i) Our model suggests that contributions that cause a campaign to succeed at the deadline must come from donations. This is a consequence of the donor valuing success and the probability of a buyer arrival in the last period being small. We empirically test this by considering all campaigns that succeed at the deadline and subtracting any last-minute donations. We find that 78 percent of campaigns would have failed if not for last-minute donations.
- (ii) A consequence of the PT equilibrium structure is that donations are made to ensure that the next buyer pledges. Hence, a donor will never donate to bring cumulative donations beyond $G - (N + 1)p$. We test this by calculating the percentage of revenue coming from buyers in the period in which a campaign succeeds. We find that the median percentage of revenues from buyers is 91 percent (see histogram in Figure D2 in online Appendix D).
- (iii) When beliefs about the probability of success are sufficiently high, donations are not required. As a result, in PT equilibria, the donation threshold drops immediately after a pledge. To explore how donations respond to contributions over time, we specify dynamic panel regressions where the dependent variable is observed donations. We use lagged values of cumulative revenues over the goal amount relative to the median campaign (within that category, time period) as controls, that is, $\mathbf{1}\{R_{j,t-k}/G_j > \text{median}[R_{\cdot,t-k}/G_{\cdot}]\}$, for $k = 1, 2, 3, 4$. We generally find a negative association of donations and progress toward the goal (for all lags), which we report in Table D2 in online Appendix D.
- (iv) The positive relationship between the donor's valuation and the time when a campaign fails arises because donors with high valuations pool with donors of lower valuations; that is, the behavior of donors of high and low valuations looks similar, but donors with higher valuations can keep campaigns active for longer. To test this prediction, we use a logistic regression to infer the failure times of campaigns, that is, $\mathbf{1}\{\text{reaches success}\}_{j,t} = x_{j,t}\beta + \varepsilon_{j,t}$,

where the outcome variable is an indicator function of the campaign outcome (success or failure) and $x_{j,t}$ includes campaign state covariates.¹² This simple prediction model produces an out-of-sample prediction accuracy of 91 percent, which we use to infer the failure time ι for each failed campaign. We define this to be the last period in which the probability of success is greater than 10 percent. We then correlate cumulative donations for failed projects with the estimated failure time. We find that the correlation is positive (corr. coef. = 0.15) and significant (t -stat = 10.1), meaning projects with larger donations fail closer to the deadline.¹³

- (v) The success-minimizing donation threshold requires donations of at least p in order to “compensate” for the absence of a buyer arrival. This motivates a simple analysis to test if campaign dynamics are consistent with success-minimizing equilibria. We calculate the average donation level for each campaign as cumulative donations (excluding the last period), divided by the number of donors observed. We label this average as μ_j^D . We also calculate the quantity-weighted reward price for each campaign, which we denote by μ_j^B . Finally, we verify if $\mu_j^D \geq \mu_j^B$, which is satisfied in success-minimizing equilibria. We find that only 29 percent of campaigns satisfy this inequality.

D. Empirical Model of Reward-Based Crowdfunding

Next, we calibrate our model to the data. Our model has three sets of unknown parameters: distribution of donor valuations, the arrival process parameters, and investor utilities. We sequentially estimate these unknowns. We outline our approach here and provide additional details in online Appendix D.

First, we estimate the distribution of donor valuations using maximum likelihood. Note that donor valuations are subject to selection. According to the model, we observe valuation realizations for campaigns that fail, but we do not observe them for campaigns that succeed.¹⁴ We account for selection using the ideas in Heckman (1979). Formally, we aggregate the data to the campaign level (j) and assume that the donor valuations w_j follow a log-linear model $\log(w_j) = x_j^W \beta^W + u_j^{(1)}$. Selection into the sample is governed by

$$s_j = \begin{cases} 1, & z_j^W \gamma^W + u_j^{(2)} > 0, \\ 0, & \text{otherwise} \end{cases}$$

where we assume a bivariate-normal distribution on the unobservables, that is, $(u_j^{(1)}, u_j^{(2)}) \sim \mathcal{N}(\mathbf{0}, (\tau^2, \tau\mu, 1))$. Hence, the unobserved components are potentially

¹² See Figure D3 in online Appendix D for more details.

¹³ Alternatively, we could infer the failure time by marking the last observed donation. However, the fact that Kickstarter promotes campaigns near the deadline likely drives some last-minute contributions. This approach implies that campaigns commonly fail at the deadline, whereas our prediction model suggests that campaigns fail much earlier.

¹⁴ More precisely, we observe only a lower bound for campaigns that succeed.

correlated. With these parametric assumptions, the log-likelihood of the donor valuation model is proportional to

$$\begin{aligned} \log(\mathcal{L}^W) \propto & \sum_j s_j \cdot \log \left\{ \Phi \left(\frac{z_j^W \gamma^W + [\log(w_j) - x_j^W \beta^W] \cdot \mu / \tau}{\sqrt{1 - \mu^2}} \right) \right\} \\ & + s_j \cdot \frac{1}{2} \frac{(\log(w_j) - x_j^W \beta^W)^2}{\tau^2} - s_j \cdot \log(\tau) \\ & + (1 - s_j) \cdot \log(\Phi(-z_j^W \gamma^W)), \end{aligned}$$

where the superscript W denotes aspects of the model that pertain to the donor and $\Phi(\cdot)$ denotes the standard normal cdf.

Second, we estimate the arrival process parameters. Without access to Kickstarter search data, buyer arrivals are also only partially observed. We observe arrivals for campaigns that succeed (as all arrivals pledge); however, we do not observe arrivals for campaigns that fail. We follow Terza (1998), who considers Poisson regression subject to sample selection. Let $n_{j,t}$ denote investor arrivals for campaign j in period t such that

$$E[n_{j,t} | x_{j,t}^A, \varepsilon_{j,t}^{(1)}] = \lambda_{j,t} = \exp(x_{j,t}^A \beta^A + \varepsilon_{j,t}^{(1)}).$$

The selection equation is given by

$$a_{j,t} = \begin{cases} 1, & z_{j,t}^A \gamma^A + \varepsilon_{j,t}^{(2)} > 0, \\ 0, & \text{otherwise} \end{cases}$$

where we again assume that the unobservables follow a bivariate-normal distribution, that is, $(\varepsilon_{j,t}^{(1)}, \varepsilon_{j,t}^{(2)}) \sim \mathcal{N}(\mathbf{0}, (\sigma^2, \sigma\rho, 1))$. With these assumptions, the log-likelihood is equal to

$$\begin{aligned} \log(\mathcal{L}^A(\theta^A)) = & \sum_j \sum_t a_{j,t} \cdot \log(\Pr(n_{j,t}, a_{j,t} = 1; x_{j,t}^A, z_{j,t}^A, \theta^A)) \\ & + (1 - a_{j,t}) \cdot \log(\Pr(a_{j,t} = 0; z_{j,t}^A, \theta^A)), \end{aligned}$$

where

$$\begin{aligned} \Pr(n_{j,t}, a_{j,t} = 1; x_{j,t}^A, z_{j,t}^A, \theta^A) \\ = \int_{-\infty}^{\infty} \frac{\lambda_{j,t}^{n_{j,t}} \cdot \exp(-\lambda_{j,t})}{n_{j,t}!} \Phi \left(\frac{z_{j,t}^A \gamma^A + \rho \varepsilon^{(1)}}{\sqrt{1 - \rho^2}} \right) \phi(\varepsilon^{(1)}) d\varepsilon^{(1)}, \end{aligned}$$

and

$$\Pr(a_{j,t} = 0; z_{j,t}^A, \theta^A) = \int_{-\infty}^{\infty} \Phi \left(\frac{z_{j,t}^A \gamma^A + \rho \varepsilon^{(1)}}{\sqrt{1 - \rho^2}} \right) \phi(\varepsilon^{(1)}) d\varepsilon^{(1)}.$$

Because the log-likelihood does not have a closed-form solution, we approximate it using the Gauss–Hermite quadrature.

Finally, we calibrate investor utilities under the additional assumption that all campaigns follow the success-maximizing equilibrium. We normalize the utility of not pledging ($v_0 = 1$), as our data do not separately identify the value of the outside option versus choosing to pledge, conditional on the other parameters. We calibrate v_j using the method of simulated moments (MSM). More precisely, for a candidate investor utility $\tilde{v}_j$, we solve for equilibrium objects, $(b(\tilde{v}_j), D_+(\tilde{v}_j), F(\tilde{v}_j))$ using the induction argument of Proposition 1. Implicitly, equilibrium objects depend on the parameters $\hat{\lambda}_j, \hat{F}_j, G_j, p_j, T_j, x_j$. After solving the dynamic game, we simulate the probability of success,

$$\Pr(\text{success} | b(\tilde{v}_j), D_+(\tilde{v}_j), F(\tilde{v}_j)) \approx \frac{1}{I} \sum_{i=1}^I \mathbf{1}\{\tilde{R}_{T_j}^{(i)} \geq G_j\},$$

where $\tilde{R}$ is the final revenue for simulation i . Success depends on strategies, beliefs, arrival realizations, and donor valuation draws. We set $I = 10,000$.

We face two challenges in calibrating v_j . First, we do not have repeat observations for a given campaign j , nor do we observe many campaigns with the exact same features (e.g., goal and reward price). Therefore, it is not possible to calibrate campaign-specific parameters. Instead, we partition the data based on campaign features. Using this partition, we match the empirical fraction of campaigns that succeed to its model counterpart; that is, for a given group of campaigns $\mathcal{G}$, we estimate $v_{\mathcal{G}}$ by solving

$$\min_{v: v - p_{\mathcal{G}} \geq 1} \left(\frac{1}{|\mathcal{G}|} \sum_{j=1}^{|\mathcal{G}|} \mathbf{1}\{\text{campaign } j \text{ succeeded}\} - \Pr(\text{success} | b(v), D_+(v), F(v)) \right)^2.$$

The second challenge is computational burden. For example, solving the game when G/p is high is difficult. For this reason, we do not calibrate parameters for all campaigns in the sample. We validate our calibration by reporting nontarget moments.

E. Empirical Specification and Parameter Estimates

For donor valuations, we include fixed effects for the campaign category, an indicator if the project is “loved” by Kickstarter, and the log of the goal amount.¹⁵ We use the log of total pledges plus 1, $\log(\sum_t B_{j,t} + 1)$, as the excluded variable that influences selection but not donor valuations. The intuition is that pledges shift the probability of success but not the distribution of donor valuations. Rather, pledges affect the donor’s strategy.

For investor arrivals, we include fixed effects for the campaign category, a quadratic expansion in time, the “loved” by Kickstarter indicator, and the log of the goal amount. We use cumulative donations divided by the goal as the excluded variable that influences selection but not arrivals. Donations shift the probability of success

¹⁵For more information on the “loved” indicator, see Figure D4 in online Appendix D.

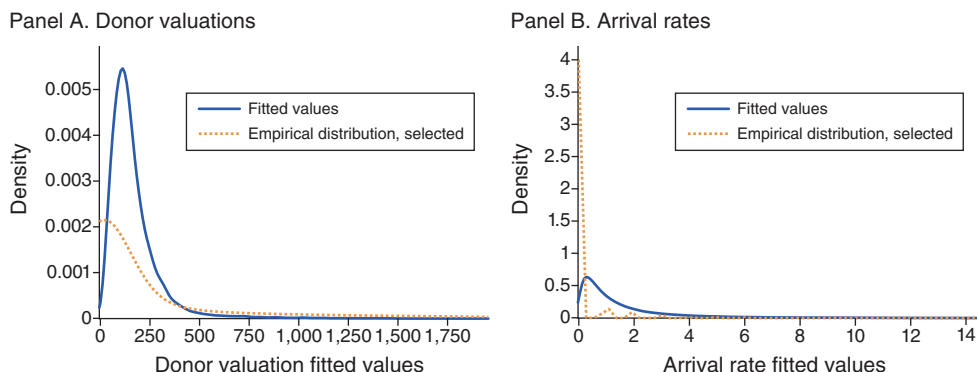


FIGURE 5. PARAMETER ESTIMATES FOR ARRIVALS AND DONORS' VALUATIONS

Notes: Panel A: Kernel density plot of the fitted values of the donor's valuation for all categories. The orange dashed line shows the empirical density of the selected sample. Panel B: Kernel density plot of the fitted values of buyer arrivals for all categories. The orange dashed line shows the empirical density of the selected sample over the range (x-axis) obtained under the model estimates up until the ninety-ninth percentile.

and investor strategies, but we assume that donations do not directly influence buyers' arrivals.

Finally, given donor valuation and arrival process parameters, investor utilities are matched to the probability of success. Intuitively, the probability of success is increasing in v : when v approaches $v_0 + p$, the probability that a campaign succeeds goes to 0, and when v increases, the probability of success can approach 1.

We limit our sample by focusing on 30-day campaigns and the first quartile of data in terms of goal amount, by campaign category. We plot fitted values for the donor's valuation distribution and buyers' arrivals in Figure 5. We provide the corresponding parameter estimates in online Appendix D. In panel A of Figure 5, we compare the fitted values for the donor's valuation to the selected sample. Accounting for sample selection shifts the distribution to the right, as we would expect. In panel B, we compare fitted arrival rates to the selected sample. For readability reasons, we exclude the long tail of empirical distribution (orange). The mean observed arrival rate is higher than the one we find when using the fitted values.

In Table A3, we provide estimates for the buyer utility from pledging, a comparison of the data moment to its model counterpart, and three sets of nontargeted moments (i.e., total donations, number of pledges, and total donations over the goal amount). We find that for some categories (e.g., fashion) the targeted and all nontargeted moments match well. However, in other categories, it is difficult to match the probability of success given our estimated arrival rates and donor valuation distribution. For example, for the theater category, our empirical estimates understate the variance in observed donations. As a consequence, we understate the probability of success compared to the one observed in the data.

In Figure 6, we provide an example equilibrium path. The initial donation spike (orange triangle at $t = 0$) is reminiscent of the role of "seed money" in charitable fundraising campaigns (Andreoni 1998). This ensures that beliefs are sufficiently high (panel B). The graphs show that beliefs jump as pledges occur. In the absence of pledges, the donor contributes to keep beliefs above the cutoff value.

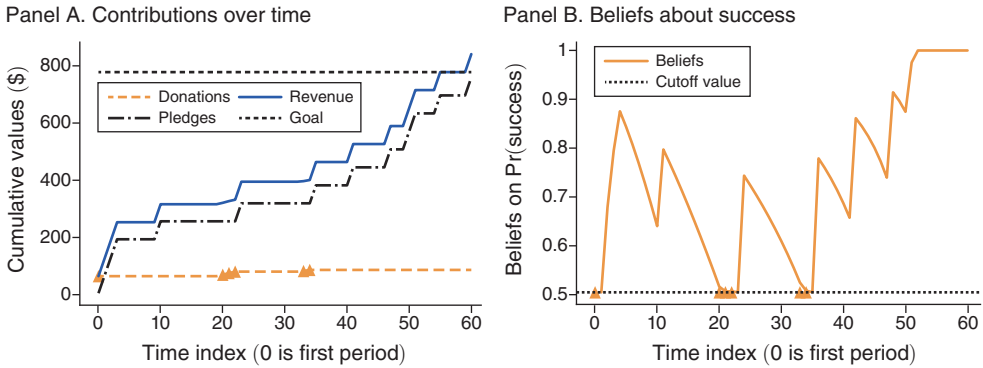


FIGURE 6. SUCCESS-MAXIMIZING EQUILIBRIUM EXAMPLE PATH

Notes: Panel A: Cumulative donations, pledges, and revenues over time for an example equilibrium path in the Journalism category, where $w = 130$. The orange triangles mark where donations occur. Panel B: Buyer beliefs about the probability of success over time. The dashed horizontal line corresponds to the cutoff value, $v_0/(v - p)$. The orange triangles mark donation events. The donor ensures that beliefs stay above the cutoff value. In this example, the campaign reaches success toward the end of the campaign.

F. Counterfactual Analysis

Using our parameter estimates, we simulate three counterfactuals: the success-minimizing equilibrium, allowing for a single donation at the beginning of the campaign, and allowing for a single donation at the end of the campaign. The latter two scenarios correspond to the alternative project designs described in Section III F.

We report counterfactual results in Table 3, focusing on the probability of success across categories. We find two key results. First, the gap in the probability of success between the success-maximizing and success-minimizing equilibrium tends to be large. We estimate the difference to be 13 percent on average, across categories. This large gap suggests that coordination benefits from dynamic signaling are high. Second, we find that the probability of success under the alternative project designs tends to be even smaller than that under the success-minimizing equilibrium. This is not obvious *ex ante*, as our theory establishes that the probability of success must be lower than that of the success-maximizing equilibrium but not necessarily lower than that of the success-minimizing equilibrium. To demonstrate the implications of this finding, we provide initial donation values under the success-maximizing equilibrium and the ones when the donor contributes only at the beginning of time across W (x-axis) in Figure 7. Panel A shows that initial donation levels when donations occur once are higher than under the success-maximizing case, but this does not lead to a higher probability of success (panel B). Allowing the donor to provide incremental support to the campaign is the best way to achieve success.

V. Conclusion

We introduce a dynamic contribution game in which randomly arriving investors can contribute to obtain an uncertain private benefit and a long-lived donor

TABLE 3—COUNTERFACTUAL RESULTS

Category	Probability of success (percent)			
	Success-maximizing	Success-minimizing	Donate at beginning	Donate at end
Art	21.8	0.4	0.3	0.0
Comics	32.0	1.1	1.0	31.2
Crafts	14.4	13.3	12.5	8.9
Design	14.6	0.3	0.3	0.0
Fashion	13.8	9.3	8.3	12.4
Film video	28.7	6.8	6.4	0.0
Food	2.3	1.1	0.9	0.9
Games	9.0	0.7	0.6	0.0
Journalism	6.6	1.6	1.5	0.0
Music	29.5	0.9	0.8	0.0
Photography	18.3	6.6	5.3	8.2
Publishing	19.9	0.5	0.5	0.0
Technology	8.3	0.0	0.0	0.0
Theater	8.8	4.2	2.9	3.2

Notes: Counterfactual estimates of the percentage of campaigns that reach the goal. We consider the success-maximizing equilibrium (assumed in estimation), success-minimizing equilibrium, and two alternative campaign designs where the donor can contribute only at the beginning or end of the campaign.

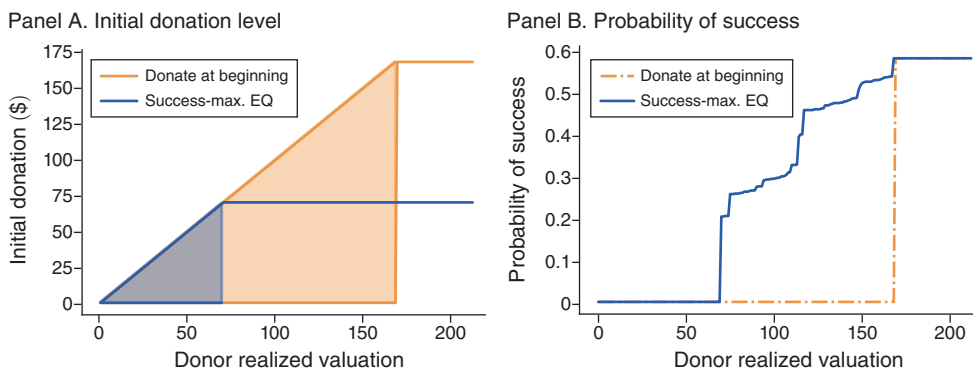


FIGURE 7. COUNTERFACTUAL EQUILIBRIUM INITIAL DONATION LEVELS AND SUCCESS

Notes: Panel A: Initial donation amount (y-axis) given realized donor valuation (x-axis) for Publishing. The shaded regions correspond to situations where there is multiplicity in initial donations because all campaigns fail. Panel B: Probability of success (y-axis) given realized valuation (x-axis). The two lines correspond to restricting the donor to contribute only at the start of the campaign (orange) and the success-maximizing equilibrium (blue).

who values the public benefits donates while seeking to minimize his total contributions required for project success. Participants face two forms of uncertainty: uncertainty in arrivals and uncertainty in the donor's valuation. We show that allowing the donor to dynamically signal his valuation by strategically timing contributions benefits all participants by facilitating coordination. Indeed, the success-minimizing equilibrium corresponds to outcomes where dynamic signaling is not possible. The success-maximizing equilibrium maximizes ex ante donor payoffs but exacerbates the uncertainty borne by investors. Investors prefer equilibria strictly in between the success-maximizing and the success-minimizing equilibria. Our model extends to many economic environments where investment is socially productive. We present

an empirical application to reward-based crowdfunding using data from Kickstarter. Our model calibrations show that restricting the donor’s ability to dynamically signal his valuation would greatly reduce the percentage of campaigns that succeed.

APPENDIX: PARAMETER ESTIMATES

TABLE A1—DONOR VALUATION PARAMETER ESTIMATES

Parameter		Valuation equation		Selection equation	
		Param. est.	Std. error	Param est.	Std. error
Corr. coef.	μ	0.708	(0.005)		
SD unobs.	τ	2.293	(0.003)	1	—
Comics		0.603	(0.037)	0.180	(0.027)
Crafts		0.047	(0.031)	0.352	(0.040)
Dance		0.841	(0.083)	−1.024	(0.049)
Design		0.704	(0.028)	0.736	(0.024)
Fashion		0.029	(0.025)	0.511	(0.022)
Film and video		0.500	(0.027)	−0.587	(0.024)
Food		0.395	(0.029)	−0.097	(0.029)
Games		0.302	(0.025)	0.647	(0.025)
Intercept		−0.880	(0.035)	−0.639	(0.038)
Journalism		−0.036	(0.057)	−0.542	(0.070)
Love		1.432	(0.050)	−0.358	(0.024)
Music		0.301	(0.027)	−0.754	(0.024)
Photography		0.255	(0.042)	0.244	(0.040)
Publishing		0.450	(0.025)	−0.174	(0.023)
Technology		0.292	(0.025)	0.601	(0.035)
Theater		0.997	(0.072)	−0.932	(0.052)
$\log(G)$		0.324	(0.004)	0.952	(0.016)
$\log(B)$				−0.993	(0.016)
Number of parameters		37			
Number of obs.		13,863			
log likelihood		−17,321.862			

Notes: Parameter estimates of donor valuations accounting for sample selection. We assume a bivariate normal distribution on the unobserved components. Standard errors computed using 1,000 bootstrap samples.

TABLE A2—INVESTOR ARRIVAL PROCESS PARAMETER ESTIMATES

Parameter		Arrival equation		Selection equation	
		Param. est.	Std. error	Param. est.	Std. error
Corr. coef.	$\operatorname{atanh}(\rho)$	−0.016	0.001		
Std. dev. unobs.	$\log(\sigma)$	0.057	0.000		
Comics		0.653	0.002	0.248	0.003
Crafts		0.050	0.005	−0.531	0.004
Dance		−0.200	0.004	0.476	0.005
Design		0.101	0.003	−0.131	0.003
Fashion		−0.110	0.002	−0.214	0.002
Film and video		−0.196	0.002	0.044	0.002
Food		−0.123	0.002	−0.168	0.003
Games		0.359	0.003	−0.453	0.003
Intercept		−6.155	0.004	−0.847	0.003
Journalism		−0.034	0.005	−0.347	0.006
Love		0.295	0.001	1.244	0.002
Music		0.114	0.002	0.367	0.002
Photography		0.106	0.003	−0.013	0.004
Publishing		0.116	0.002	−0.057	0.002
Technology		−0.279	0.003	−0.528	0.003
Theater		−0.016	0.003	0.556	0.004
Time-linear		−0.190	0.000	0.031	0.000
Time-quadratic		0.007	0.000	−0.000	0.000
$\log(G)$		0.730	0.001	−0.117	0.000
$\log(D)$				9.474	0.014
Corr. coef.	ρ	−0.016	0.001		
Std. dev. unobs.	σ	1.057	0.034	1	—
Number of parameters		41			
Number of observations		845,643			
log likelihood		−385,554.946			

Notes: Parameter estimates of arrival process accounting for sample selection. We assume a bivariate normal distribution on the unobserved components. Standard errors computed using 1,000 bootstrap samples.

TABLE A3—PARAMETER ESTIMATES FOR INVESTOR UTILITY

Category	Nontargeted moments (data, model)								
	Util. est.	Data	Model	D_T		N_T		D_T/G	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Art	62.440	0.215	0.217	41.766	38.346	5.526	2.481	0.100	0.088
Comics	50.167	0.284	0.320	79.629	60.320	11.968	10.430	0.100	0.081
Crafts	42.336	0.130	0.144	45.877	37.593	4.024	3.369	0.093	0.084
Design	85.610	0.147	0.146	117.677	119.530	11.639	4.010	0.078	0.074
Fashion	61.113	0.138	0.138	40.783	43.667	5.619	5.854	0.063	0.055
Film and video	95.442	0.289	0.287	192.842	83.390	6.142	4.642	0.175	0.079
Food	124.876	0.185	0.023	233.265	103.496	8.432	8.786	0.113	0.045
Games	46.697	0.091	0.090	61.048	75.074	9.300	5.604	0.054	0.056
Journalism	65.248	0.065	0.066	53.058	49.135	1.419	1.198	0.064	0.063
Music	60.414	0.300	0.295	195.941	63.952	7.281	7.198	0.182	0.065
Photography	50.367	0.183	0.183	96.783	61.013	5.662	9.288	0.109	0.074
Publishing	53.491	0.201	0.199	101.197	77.380	5.727	4.892	0.109	0.088
Technology	112.169	0.084	0.083	156.271	93.679	7.773	3.390	0.058	0.033
Theater	68.407	0.422	0.088	398.868	167.459	8.000	12.860	0.273	0.126

Notes: Calibration of buyer utilities assuming the success-maximizing equilibrium and using donor valuation estimated in Table A1 and arrival process estimated in Table A2. We consider each category separately and match moments for the first quartile of campaigns in terms of goal amount. We set $v_0 = 1$. We report our targeted moment (i.e., fraction of campaigns that succeed) and nontargeted moments.

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